

```
# import python libraries
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt # visualizing data
%matplotlib inline
import seaborn as sns
```

```
# import csv file
df = pd.read_csv('Diwali Sales Data.csv', encoding= 'unicode_escape')
```

```
df.shape
```

```
(11251, 15)
```

```
df.head()
```

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status
0	1002903	Sanskriti	P00125942	F	26-35	28	0
1	1000732	Kartik	P00110942	F	26-35	35	1
2	1001990	Bindu	P00118542	F	26-35	35	1
3	1001425	Sudevi	P00237842	M	0-17	16	0
4	1000588	Joni	P00057942	M	26-35	28	1

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  -
0   User_ID               11251 non-null  int64
1   Cust_name             11251 non-null  object
2   Product_ID           11251 non-null  object
3   Gender                11251 non-null  object
4   Age Group             11251 non-null  object
5   Age                   11251 non-null  int64
6   Marital_Status        11251 non-null  int64
7   State                 11251 non-null  object
8   Zone                  11251 non-null  object
9   Occupation            11251 non-null  object
10  Product_Category      11251 non-null  object
11  Orders                11251 non-null  int64
12  Amount                11239 non-null  float64
13  Status                 0 non-null      float64
14  unnamed1               0 non-null      float64
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB
```

```
#drop unrelated/blank columns
df.drop(['Status', 'unnamed1'], axis=1, inplace=True)
```

```
#check for null values  
pd.isnull(df).sum()
```

	0
User_ID	0
Cust_name	0
Product_ID	0
Gender	0
Age Group	0
Age	0
Marital_Status	0
State	0
Zone	0
Occupation	0
Product_Category	0
Orders	0
Amount	12

dtype: int64

```
# drop null values  
df.dropna(inplace=True)
```

```
# change data type  
df['Amount'] = df['Amount'].astype('int')
```

```
df['Amount'].dtypes
```

```
dtype('int64')
```

```
df.columns
```

```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group',
      'Age',
      'Marital_Status', 'State', 'Zone', 'Occupation',
      'Product_Category',
      'Orders', 'Amount'],
      dtype='object')
```

```
#rename column
```

```
df.rename(columns= {'Marital_Status':'Shaadi'})
```

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Shaadi	
0	1002903	Sanskriti	P00125942	F	26-35	28	0	
1	1000732	Kartik	P00110942	F	26-35	35	1	And
2	1001990	Bindu	P00118542	F	26-35	35	1	U
3	1001425	Sudevi	P00237842	M	0-17	16	0	
4	1000588	Joni	P00057942	M	26-35	28	1	
...	...	...	...	...	...	...	...	
11246	1000695	Manning	P00296942	M	18-25	19	1	
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	
11248	1001209	Oshin	P00201342	F	36-45	40	0	
11249	1004023	Noonan	P00059442	M	36-45	37	0	

```
# describe() method returns description of the data in the DataFrame
df.describe()
```

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

```
# use describe() for specific columns
df[['Age', 'Orders', 'Amount']].describe()
```

	Age	Orders	Amount
count	11239.000000	11239.000000	11239.000000
mean	35.410357	2.489634	9453.610553
std	12.753866	1.114967	5222.355168
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	2.000000	8109.000000
75%	43.000000	3.000000	12675.000000
max	92.000000	4.000000	23952.000000



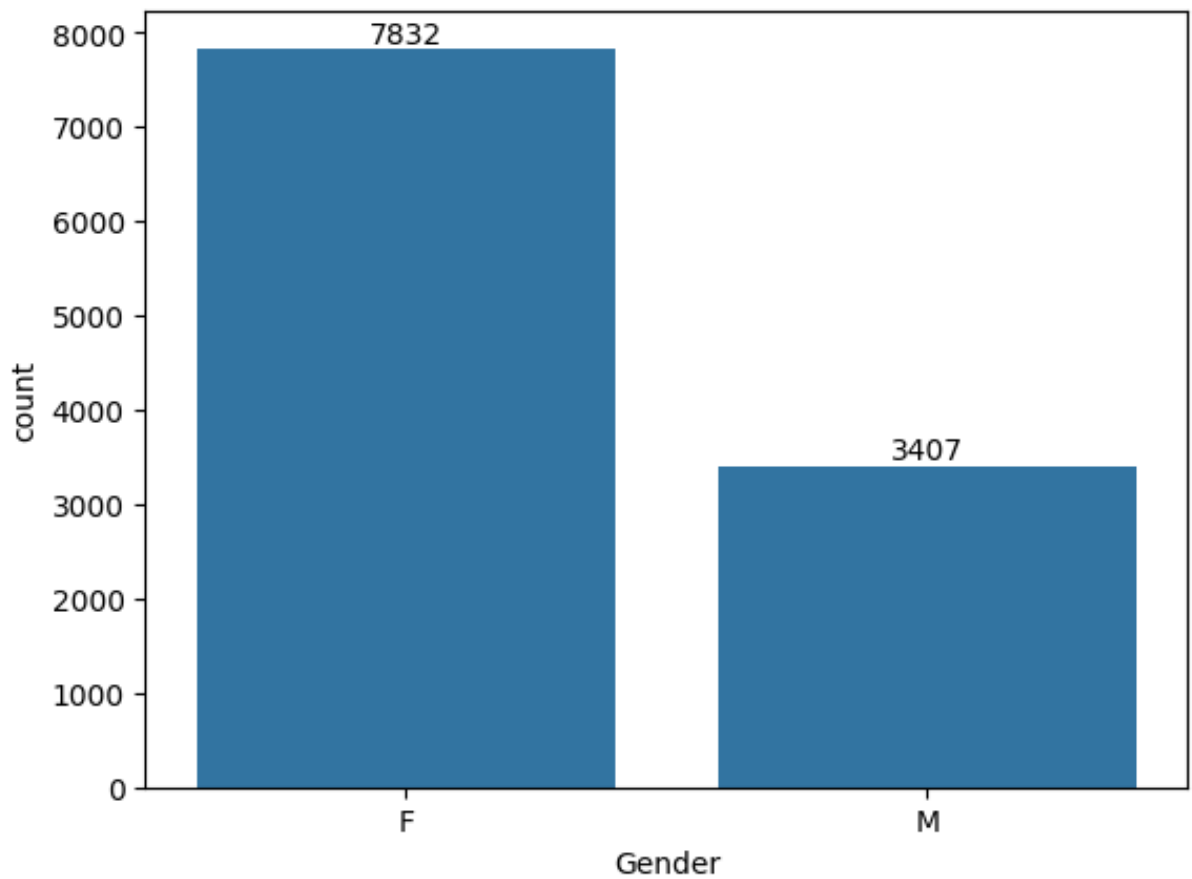
Exploratory Data Analysis

By Gender

```
# plotting a bar chart for Gender and it's count
```

```
ax = sns.countplot(x = 'Gender',data = df)
```

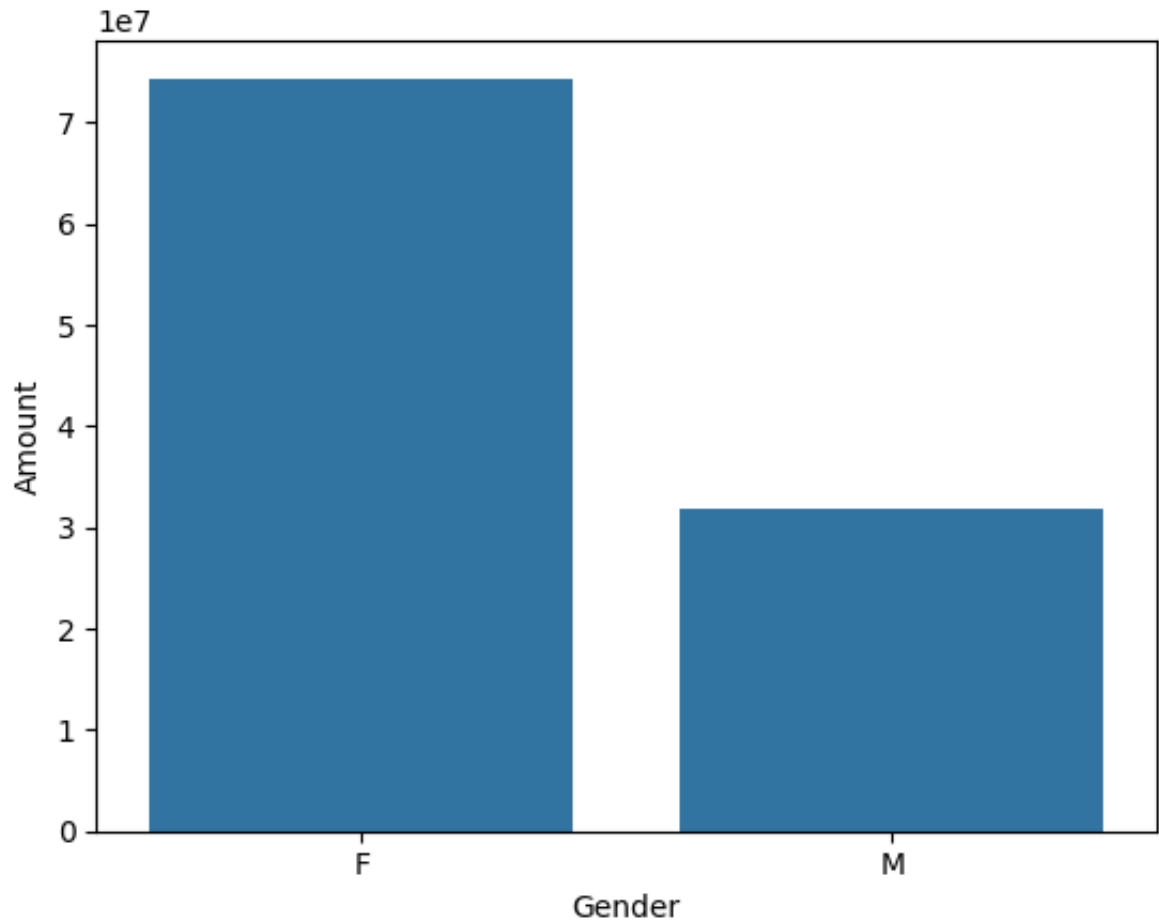
```
for bars in ax.containers:  
    ax.bar_label(bars)
```



```
# plotting a bar chart for gender vs total amount
```

```
sales_gen = df.groupby(['Gender'], as_index=False)['Amount'].sum().s  
sns.barplot(x = 'Gender',y= 'Amount' ,data = sales_gen)
```

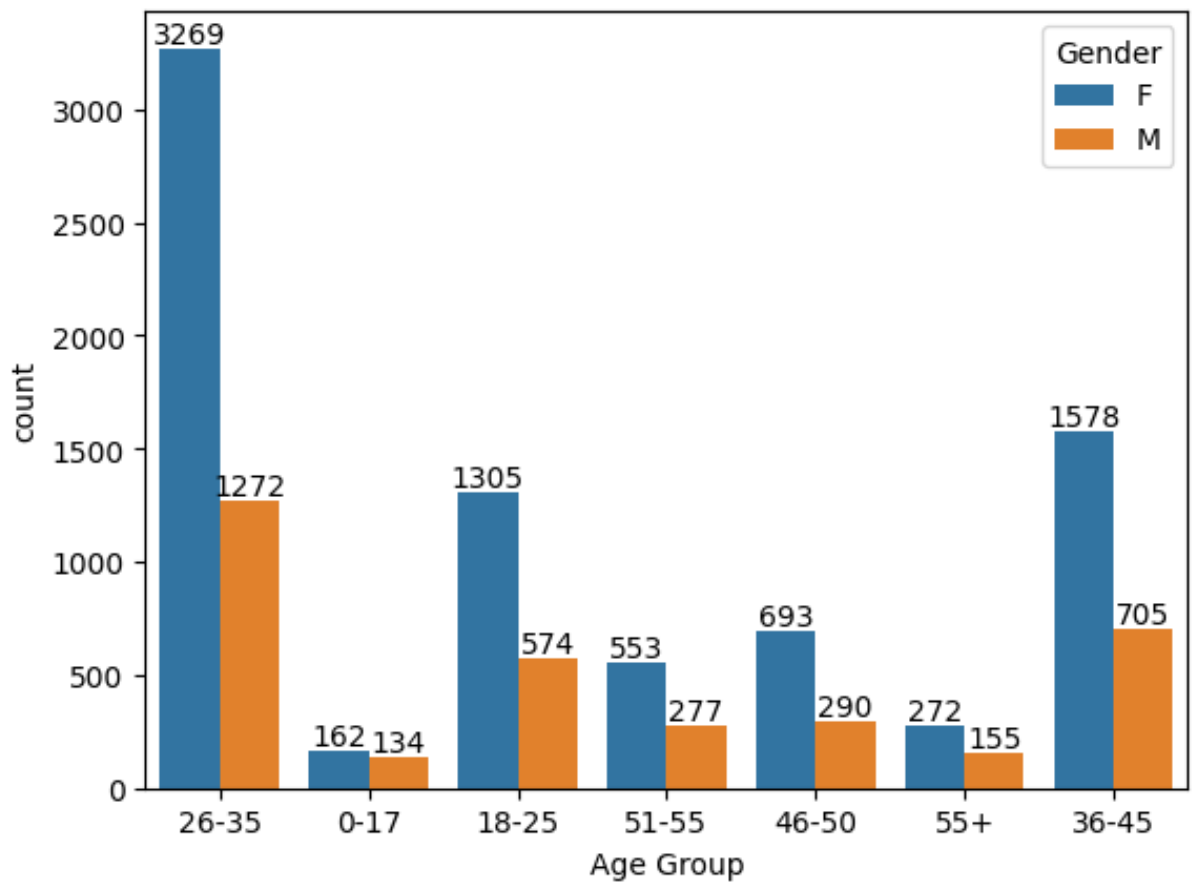
```
<Axes: xlabel='Gender', ylabel='Amount'>
```



From above graphs we can see that most of the buyers are females and even the purchasing power of females are greater than men

Age

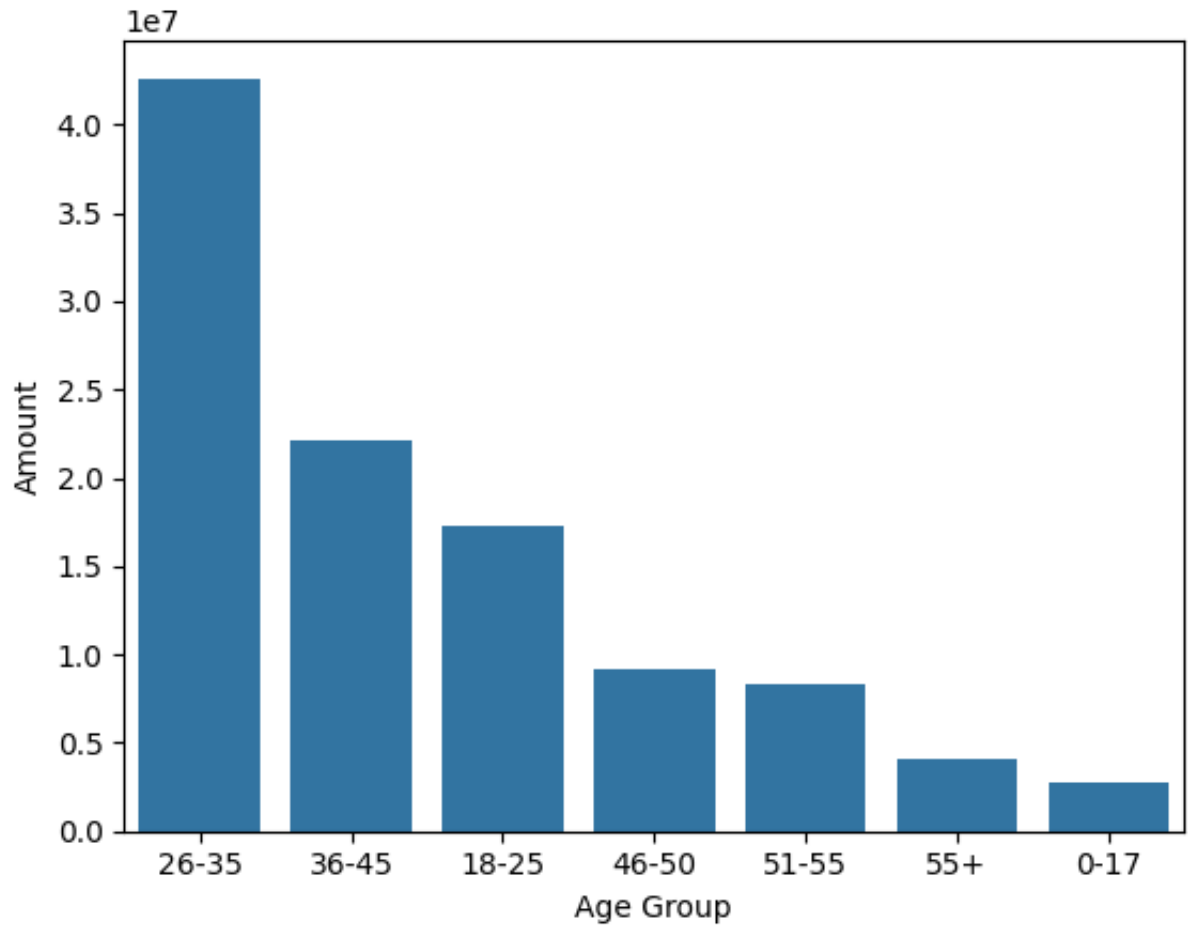
```
ax = sns.countplot(data = df, x = 'Age Group', hue = 'Gender')  
  
for bars in ax.containers:  
    ax.bar_label(bars)
```




```
# Total Amount vs Age Group
sales_age = df.groupby(['Age Group'], as_index=False)['Amount'].sum()

sns.barplot(x = 'Age Group',y= 'Amount' ,data = sales_age)
```

```
<Axes: xlabel='Age Group', ylabel='Amount'>
```



From above graphs we can see that most of the buyers are of age group between 26-35 yrs female

State

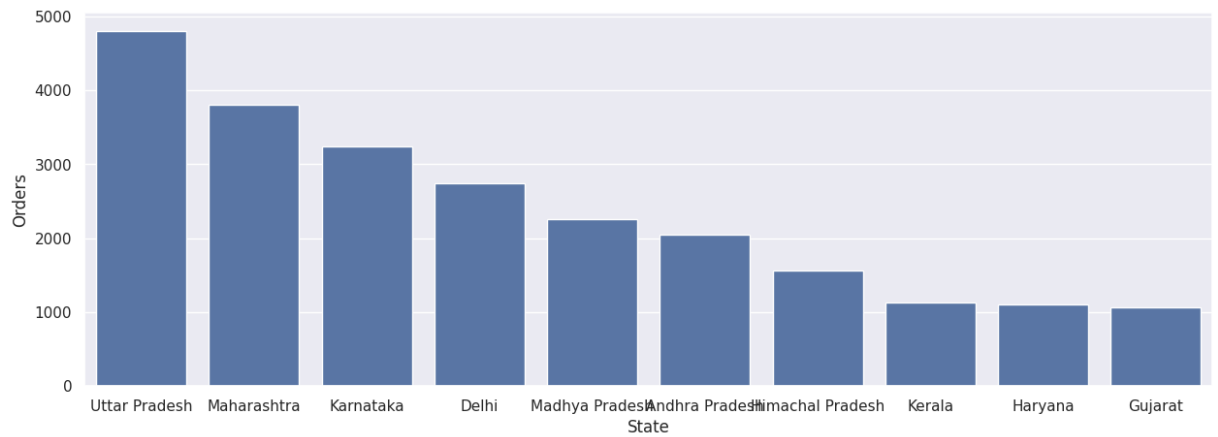
```
# total number of orders from top 10 states
```

```
sales_state = df.groupby(['State'], as_index=False)['Orders'].sum()
```

```
sns.set(rc={'figure.figsize':(15,5)})
```

```
sns.barplot(data = sales_state, x = 'State',y= 'Orders')
```

<Axes: xlabel='State', ylabel='Orders'>

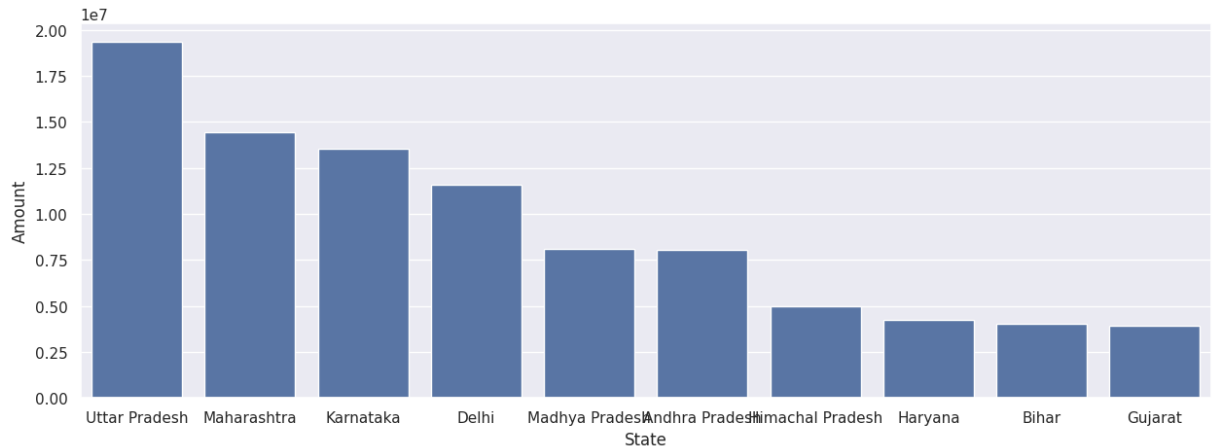


```
# total amount/sales from top 10 states

sales_state = df.groupby(['State'], as_index=False)['Amount'].sum()

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State', y= 'Amount')
```

<Axes: xlabel='State', ylabel='Amount'>

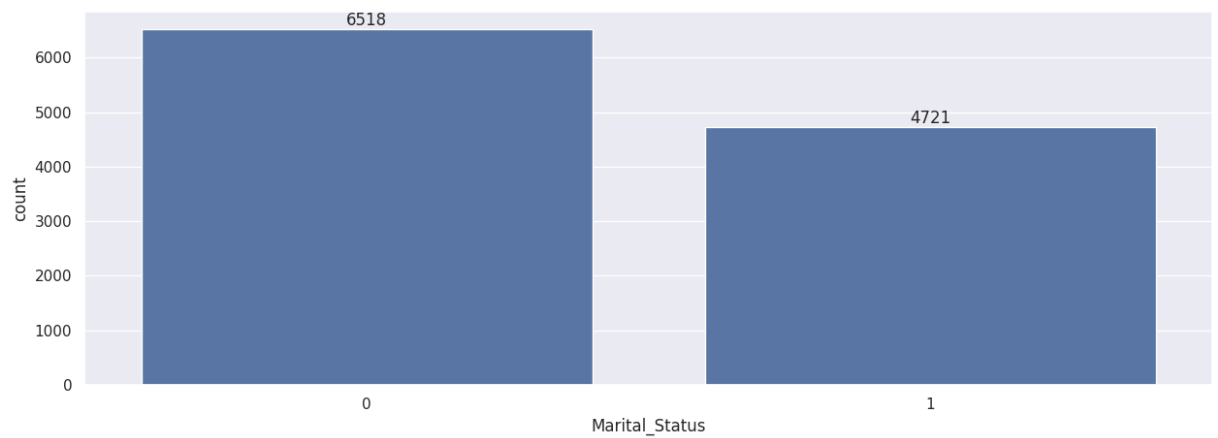


From above graphs we can see that most of the orders & total sales/amount are from Uttar Pradesh, Maharashtra and Karnataka respectively

Marital Status

```
ax = sns.countplot(data = df, x = 'Marital_Status')

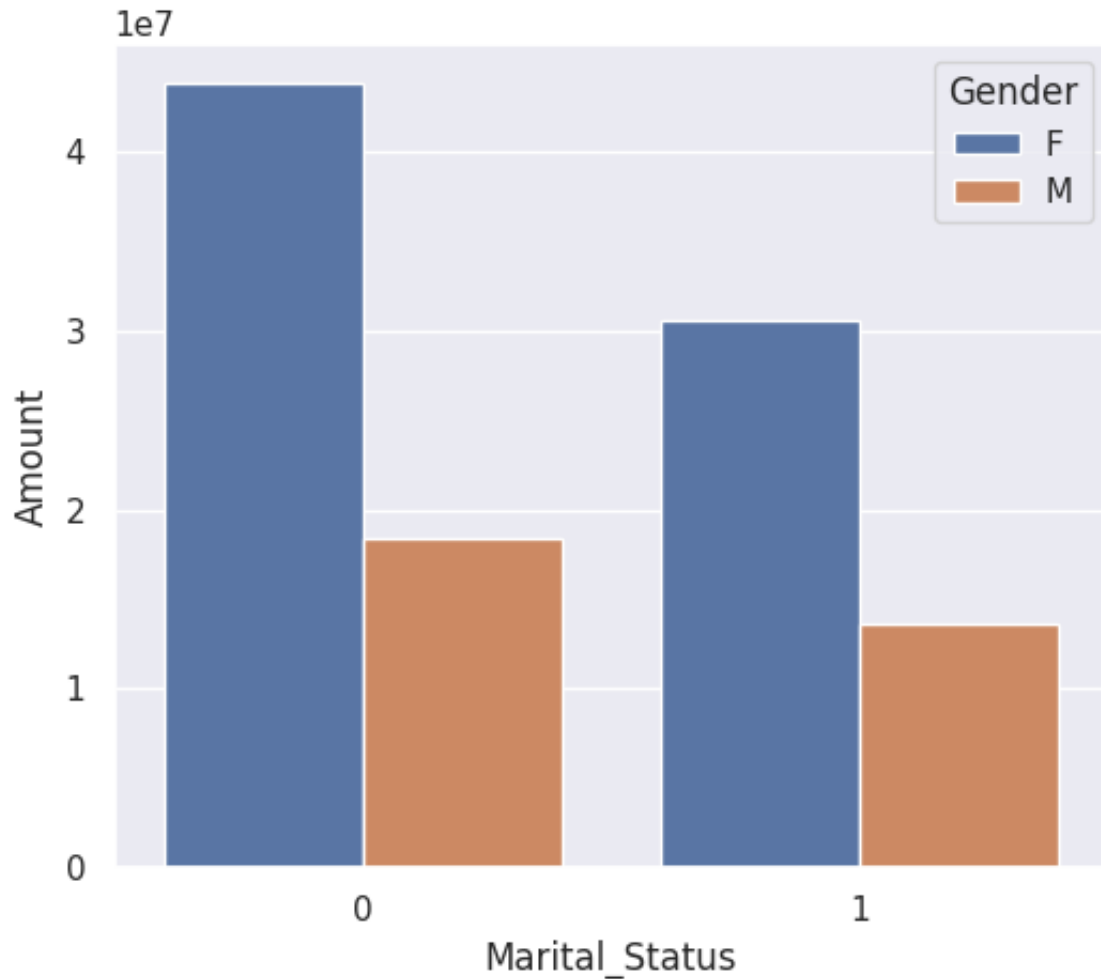
sns.set(rc={'figure.figsize':(7,5)})
for bars in ax.containers:
    ax.bar_label(bars)
```



```
sales_state = df.groupby(['Marital_Status', 'Gender'], as_index=False)

sns.set(rc={'figure.figsize':(6,5)})
sns.barplot(data = sales_state, x = 'Marital_Status', y= 'Amount', hue = 'Gender')
```

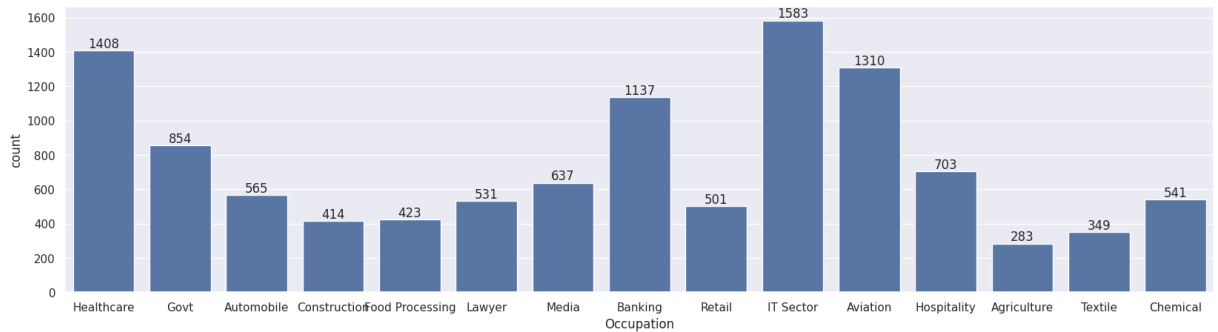
<Axes: xlabel='Marital_Status', ylabel='Amount'>



From above graphs we can see that most of the buyers are married (women) and they have high purchasing power

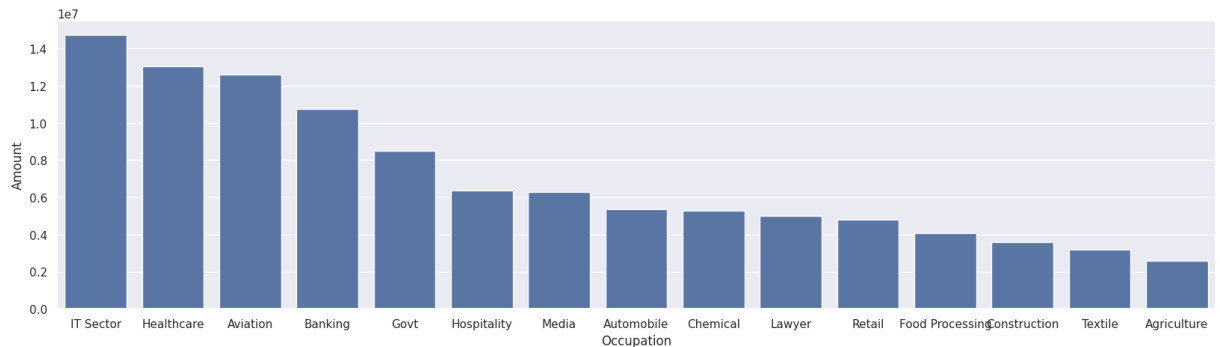
Occupation

```
sns.set(rc={'figure.figsize':(20,5)})  
ax = sns.countplot(data = df, x = 'Occupation')  
  
for bars in ax.containers:  
    ax.bar_label(bars)
```



```
sales_state = df.groupby(['Occupation'], as_index=False)['Amount'].s  
  
sns.set(rc={'figure.figsize':(20,5)})  
sns.barplot(data = sales_state, x = 'Occupation',y= 'Amount')
```

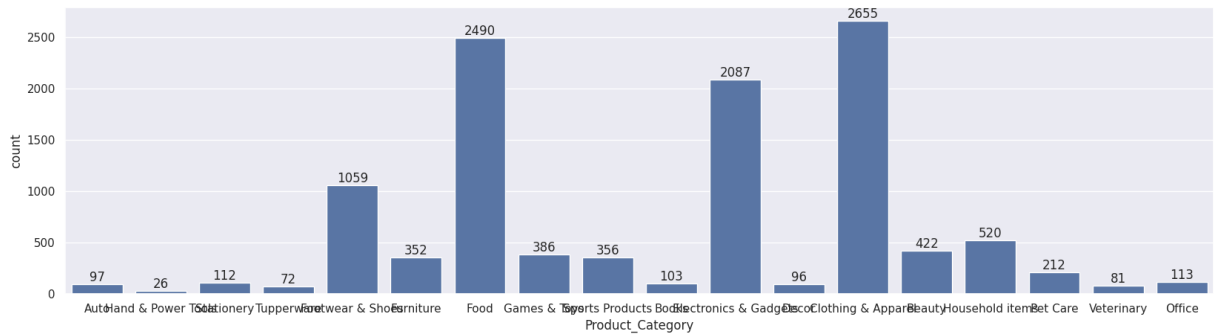
<Axes: xlabel='Occupation', ylabel='Amount'>



From above graphs we can see that most of the buyers are working in IT, Healthcare and Aviation sector

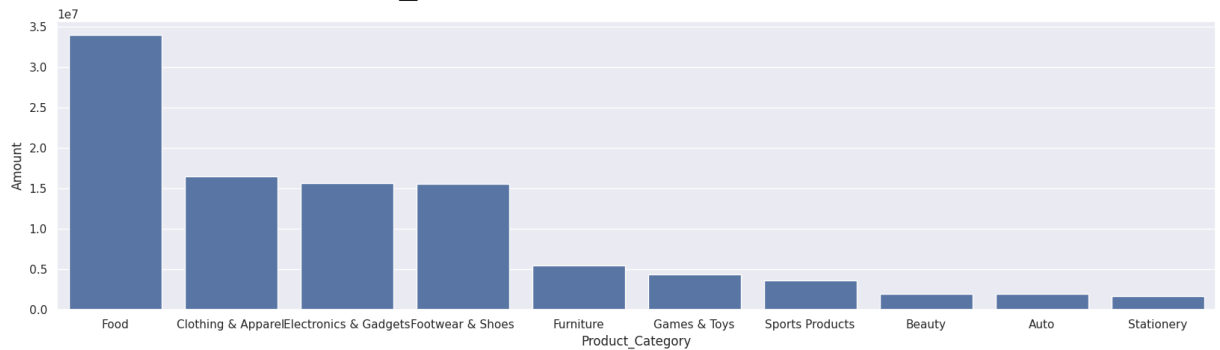
Product Category

```
sns.set(rc={'figure.figsize':(20,5)})  
ax = sns.countplot(data = df, x = 'Product_Category')  
  
for bars in ax.containers:  
    ax.bar_label(bars)
```




```
sales_state = df.groupby(['Product_Category'], as_index=False)['Amount']  
  
sns.set(rc={'figure.figsize':(20,5)})  
sns.barplot(data = sales_state, x = 'Product_Category',y= 'Amount')
```

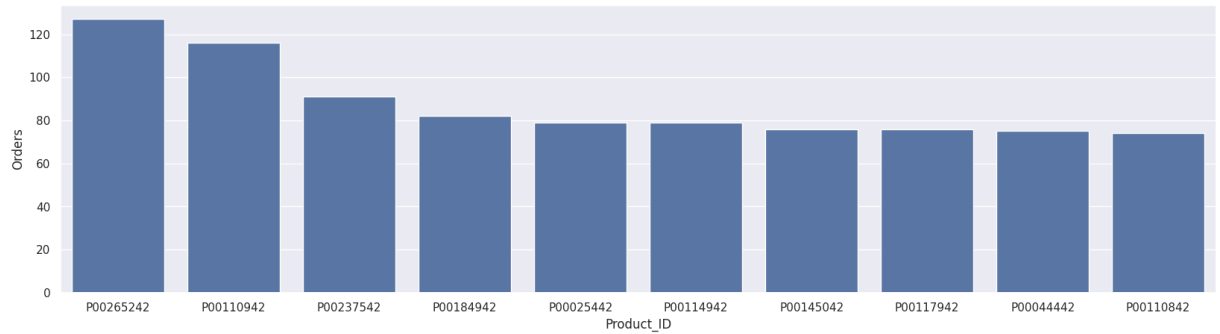
<Axes: xlabel='Product_Category', ylabel='Amount'>



From above graphs we can see that most of the sold products are from Food, Clothing and Electronics category

```
sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].s  
  
sns.set(rc={'figure.figsize':(20,5)})  
sns.barplot(data = sales_state, x = 'Product_ID',y= 'Orders')
```

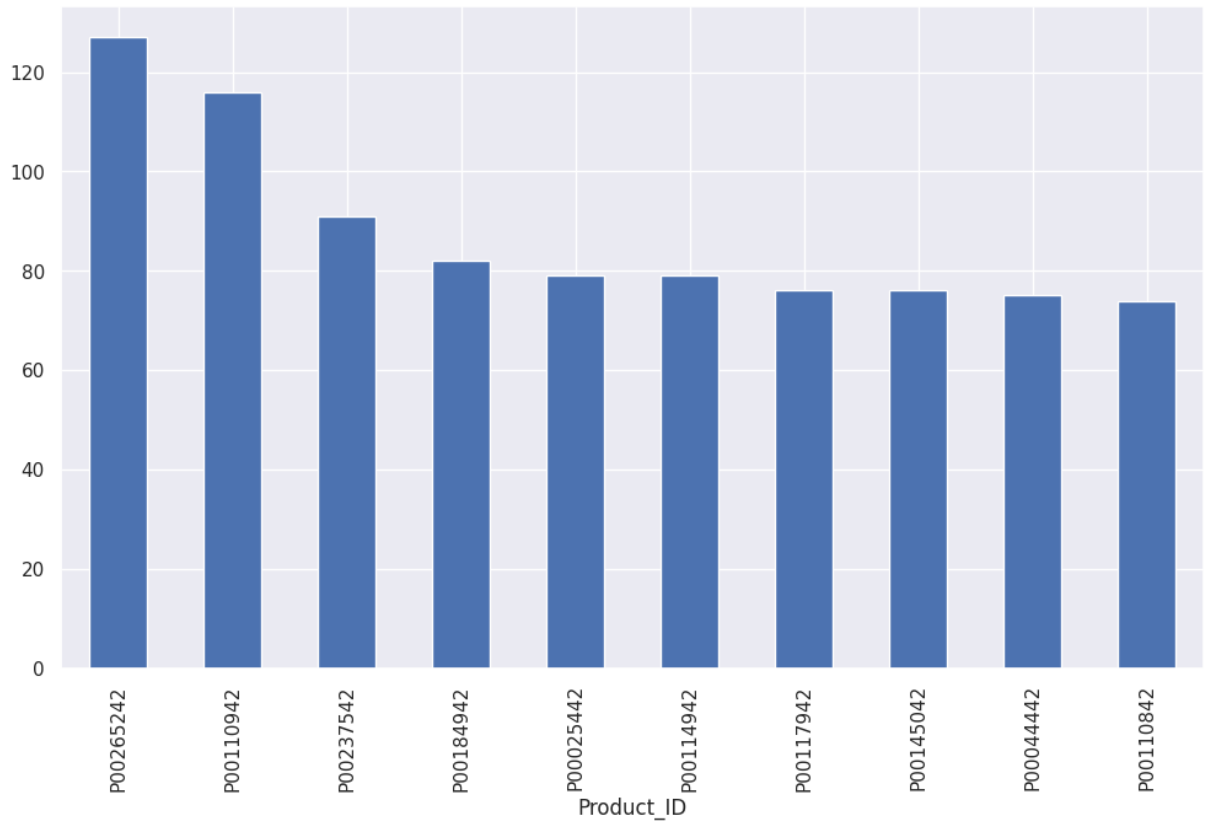
<Axes: xlabel='Product_ID', ylabel='Orders'>



```
# top 10 most sold products (same thing as above)
```

```
fig1, ax1 = plt.subplots(figsize=(12,7))  
df.groupby('Product_ID')['Orders'].sum().nlargest(10).sort_values(a
```

```
<Axes: xlabel='Product_ID'>
```



Conclusion:

Married women age group 26-35 yrs from UP, Maharastra and Karnataka working in IT, Healthcare and Aviation are more likely to buy products from Food, Clothing and Electronics category